

The following notes and instruction set is to be used when you notice that AIMS is running excessively slow.

During your investigation you find that the sqlservr.exe process is using an excessive amount of memory. Anything above 2GB of RAM is bordering on excessive.

The result of the investigation may push you to forcing SQL into Checking Mode (aka, "Checkpointing").

What is Checkpointing and why is it necessary?

- As SQL runs data being written permanently to tables is delayed based on resource. SQL uses cached disk blocks to temporarily store the information, writing it permanently to table at a later time when resource is more available.
- Checkpointing is an internal process that writes all dirty pages (modified pages) from Buffer Cache to Physical disk, apart from this it also writes the log records from log buffer to physical file. Writing of Dirty pages from buffer cache to data file is also known as Hardening of dirty pages.
 - It is a dedicated process and runs automatically by SQL Server at specific intervals. SQL Server runs checkpoint process for each Database individually.
- Checkpoint helps to reduce the recovery time for SQL Server in the event of unexpected shutdown or system crash\Failure.

In SQL Server 2017 there are four types of Checkpoints:

Automatic: This is the most common checkpoint which runs as a process in the background to make sure SQL Server Database can be recovered in the time limit defined by the Recovery Interval – Server Configuration Option.

Indirect: This is new in SQL Server 2012 and beyond. This also runs in the background but to meet a user-specified target recovery time for the specific Database where the option has been configured. Once the Target_Recovery_Time for a given database has been selected this will override the Recovery Interval specified for the server and avoid Automatic Checkpoint on such DB.

Manual: This one runs just like any other T-SQL statement, once you issue checkpoint command it will run to its completion. Manual Checkpoint runs for your current Database Only. You can also specify the Checkpoint_Duration which is optional, this duration specifies the time in which you want your checkpoint to complete.

Internal: As a user you can't control Internal Checkpoint. Issued on specific operations such as:

1. Shutdown initiates a Checkpoint operation on all databases except when Shutdown is not clean (Shutdown with nowait)
2. If the recovery model gets changed from Full\Bulk-logged to Simple.
3. While taking Backup of the Database.
4. If your DB is in Simple Recovery model, checkpoint process executes automatically either when the log becomes 70% full, or based on Server option-Recovery Interval.
5. Alter Database command to add or remove a data\log file also initiates a checkpoint.
6. Checkpoint also takes place when the recovery model of the DB is Bulk-Logged and a minimally logged operation is performed.
7. DB Snapshot creation.

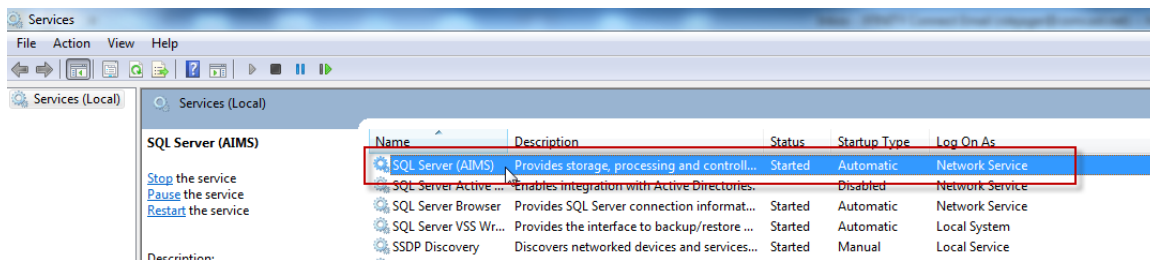
Checkpointing is the act of flushing modified, cached database blocks to disk.

Steps for AIMS SQL Checkpointing:

Make sure customer is not running or processing any jobs.

Open up Windows Services Screen, and stop the NeopostIO, NeopostEnterprise Services (if your AIMS is older you will need to also stop NeopostAudit).

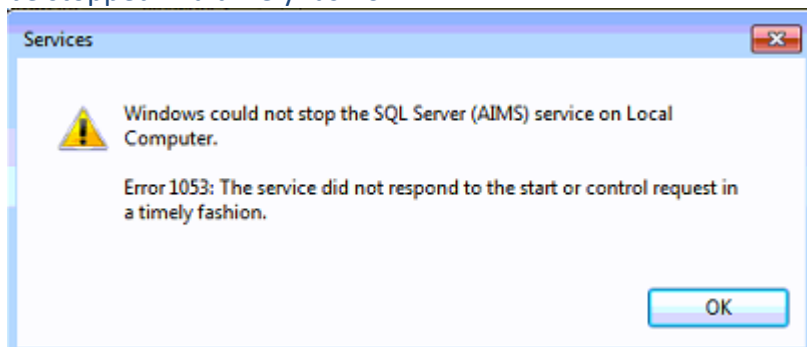
Then find the SQL AIMS service.



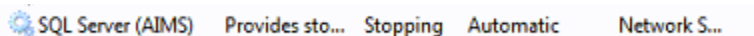
Right Click and select properties.

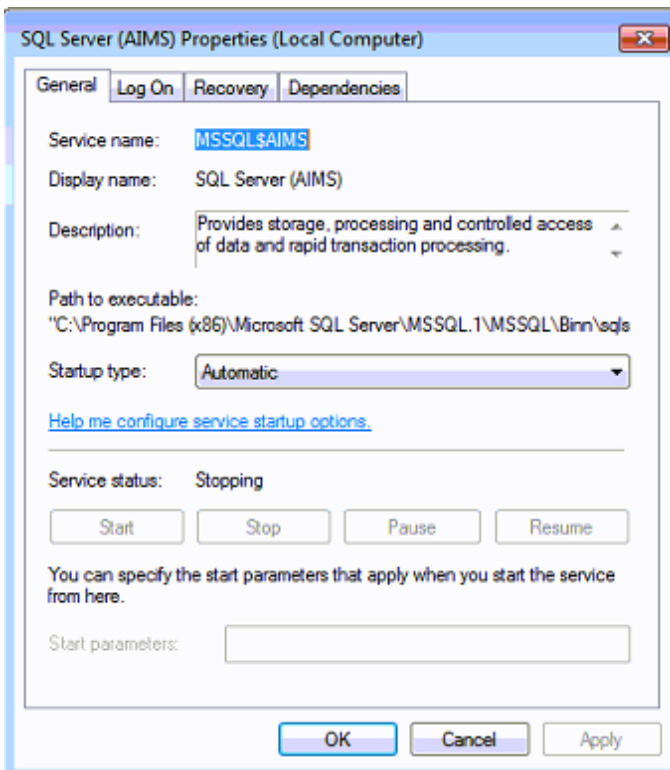
Manually stop the service.

Keep track on how long it takes to stop, you might be given an error message stating that the service cannot be stopped in a timely fashion.



Click "OK" if you get this dialog; it will continue trying to stop the service.





You may need to refresh the page a few times until you see the service stop.

Once it has stopped can you the manually restart the service and again keep a track on how long it takes to restart.

After the service is up and running again restart the PC.

Just restarting the PC will not restart this service as the PC shuts down quicker than the service will restart.

Once the PC has booted back up check the memory usage and hopefully the slowdown will have cleared.

The reason you are asked to track the time to Stop and Start the service - is because it gives us an idea of whether SQL has been successfully "check pointing" the databases.

This check pointing should occur automatically on a regular basis and on backups and shutting down of SQL Server.

There is an issue we have come across on a couple of other sites where the customer shuts down their AIMS PC a lot and the SQL Server was not being given enough time to checkpoint the databases on PC shutdown. If SQL Server service takes a long time to shutdown this is because it is checkpointing the databases.